

Gerald Ritson

Senior Software Engineer

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SUMMARY

Senior Software Engineer with over 11 years of expertise in developing and enhancing web and mobile applications. Skilled in steering projects from concept through deployment, ensuring seamless user interactions and peak performance. Dedicated to continuous professional development and innovation, consistently seeking innovative approaches to enhance project outcomes and exceed client expectations.

EDUCATION

Bachelor of Science in Computer Science

University at Buffalo

2010 - 2014

SKILLS

JavaScript, TypeScript, Node.js, Express.js, Nest.js, Python, Django, Flask, FastAPI, Go, React, Next.js, Vue.js, React Native, Redux, Mobx, RxJs, NgRx, MUI, Tailwind CSS, Bootstrap, SASS, LESS, PostgreSQL, MySQL, NoSQL, MongoDB, Snowflake, Cassandra, DynamoDB, Redis, Neo4j, Docker, Kubernetes, Apache Kafka, RabbitMQ, gRPC, Elasticsearch, Jenkins, GitHub, SQL, Cypher, GraphQL, GitLab, Git, Linux, Cypress, PyUnit, Pytest, Karma, Mocha, Jest, Agile, Scrum, TDD, BDD

EXPERIENCE

Senior Software Engineer

09/2023 - Present

Solstice

- Led the development of an innovative system for robotic delivery, featuring multi-order optimization and real-time tracking
- Designed and developed microservices-based applications using **Node.js** frameworks like Nest.js, and deployed them on AWS ECS or AWS EKS
- Utilized **TypeORM** for object-relational mapping, facilitating data access and manipulation while maintaining high performance
- Wrote complex queries and performed data analysis with **MongoDB**, optimizing data retrieval processes and supporting business intelligence initiatives
- Spearheaded the design and development of real-time analytics platform using **Python**, Vue.js, AWS IoT Core, Kafka, and Grafana, resulting 40% improvement in the operational insights
- Integrated custom middleware, implemented dependency injection systems, and established security protocols within **FastAPI** to improve the robustness and flexibility of APIs
- Developed data visualizations using Python libraries such as **Matplotlib** and integrated **D3.js**, improving decision-making
- Worked with ML engineers to implement an efficient **Python** algorithm for optimizing delivery routes by batching orders with similar destinations
- Played a key role the implementation of advanced monitoring solutions and established best practices for **Elasticsearch**, while leveraging AWS Lambda to execute code without managing servers
- Designed and implemented efficient caching strategies using **Redis** to enhance application performance
- Implemented DevOps tools such as Jenkins, Docker, **Kubernetes**, and Terraform, streamlining workflows and enhancing release velocity by 40% while improving deployment stability
- Collaborated with **Snowflake** cloud data warehouse and AWS S3 to integrate data from multiple source systems, including loading nested JSON formatted data into Snowflake tables
- Developed robust CLI tools in **Golang** to enhance development workflows and automate routine tasks
- Conducted thorough code reviews and mentored developers through pair programming sessions to uphold high-quality standards

EXPERIENCE

Senior Full Stack Engineer

05/2020 - 09/2023

Codup

- Designed and developed scalable, low-latency e-commerce applications using **Node.js**, Python, and Vue, efficiently handling over 300K concurrent users with optimized performance
- Developed and maintained RESTful and **GraphQL** web services with Node.js, utilizing NoSQL databases like Apache Cassandra and MongoDB for handling large datasets
- Implemented real-time dashboards using Logstash, **Elasticsearch**, Kibana, and **Redis**, with Vue.js and **FastAPI**
- Developed dynamic graphical reports using **NumPy** and Matplotlib
- Utilized **Celery** workers to schedule tasks and perform cron jobs based on dependencies and upstream jobs, resolving conflicts in resource allocation
- Employed the **python-twitter** library to retrieve Twitter feeds based on keyword and used Beautiful Soup for web scraping to extract data for building graphs
- Leveraged **PyUnit** for testing and debugging Python applications, achieving 95% test coverage
- Integrated **Okta** across multiple applications using Python SDKs, implementing SAML and OpenID Connect
- Developed and optimized ETL pipelines using **PySpark** on Azure Databricks, for large-scale data processing
- Developed Python scripts for task automation, managing dependencies, and implementing SLA watchers and time sensors using Apache Airflow for efficient workflow management
- Developed data mappings using **Python** and **Pandas** to clean and populate staging tables, transformed data to meet business needs within the Enterprise Data Warehouse, and populated the DataMart with essential information
- Integrated **Pytest** with CI/CD pipelines to automate test execution and reporting, ensuring consistent code quality throughout the development lifecycle
- Managed MySQL fail-over in auto mode, facilitating automatic switchover from primary to replica servers
- Designed and implemented data models using relational databases such as **PostgreSQL** and MySQL, while optimizing NoSQL database queries to enhance performance

Full Stack Engineer

01/2016 - 05/2020

Grawlix

- Built and maintained several betting platforms using Node.js, and React.js
- Developed casino-style games including **Poker**, **Blackjack**, and **Roulette**, using React.js, JavaScript, and Tailwind CSS
- Built RESTful APIs using **Express.js**, and deploying them on AWS API Gateway for secure and managed API endpoints
- Implemented test cases with **Jest** and **Mocha**
- Designed front-end and mobile applications for user-interactive web pages using technologies such as TypeScript, React.js, and **React Native** with Redux Saga
- Utilized React.js and **Redux** to develop isomorphic applications, leveraging GraphQL queries and the **Apollo GraphQL** library for seamless data management and efficient server-side rendering
- Implemented **Redux Saga** with actions and reducers to perform asynchronous API calls
- Developed a POC to pick an optimal state management tool between Redux and **Mobx**
- Developed, linted, and managed **Docker** files and containers, including configuring volumes, managing container-based databases, and optimizing containerized services
- Integrated **OAuth 2.0** to manage resource access with access tokens through an authorization server and resource server setup
- Deployed the systems on **AWS** infrastructure, using services like DynamoDB, **Lambda**, CloudWatch, EC2, S3, and SQS
- Created a Lambda Deployment function to automatically handle event triggers from the **S3** bucket
- Migrated PostgreSQL and **MySQL** databases from on-premises to AWS EC2 and RDS environments
- Leveraged REST tools, including Swagger, Postman, and REST/SOAP UI, for testing HTTP requests and responses to ensure API functionality and reliability
- Collaborated with an Agile team, routinely applying **Agile** methodologies and participating in regular Scrum meetings led by the Scrum Master

EXPERIENCE

Frontend Developer

05/2014 - 01/2016

XDuce

- Worked with a cross-functional team to develop high-traffic e-commerce web applications using **AngularJS** 1.x, seamlessly integrating with a PHP CodeIgniter backend
- Developed modular and reusable UI components in **AngularJS** with directives and services
- Spearheaded enhancements to website performance by utilizing AngularJS features to reduce DOM manipulation and improve load times, increasing user engagement by 20%
- Translated complex **PSD** web designs into pixel-perfect, responsive HTML templates using HTML5, CSS3, JavaScript, and **Bootstrap**, ensuring accuracy in design and functionality across all major browsers and devices
- Applied advanced CSS techniques, including **Flexbox** and Grid, to create fluid layouts from static designs, improving user experience on both desktop and mobile platforms
- Optimized web applications using responsive design techniques to ensure functionality across both full and mobile screen sizes, and implemented best practices for **SEO** in webpage design
- Employed version control systems like **Git** to manage and track changes in frontend codebases